utility10_02-gas-taraju

Gas Taraju (LPG cylinder scale): manufacturing manual

> **Status: design only.** Nothing here has been built or tested as hardware. This manual tells you exactly what to make, buy, assemble, program and test, and how to accept or reject a unit; expect to correct it after the first prototype run (5 units).

Product summary. The cylinder stands on a 330 mm plate over four load cells. After the cylinder is placed and still for 3 s the scale subtracts the tare stamped on the collar and shows % and kg left. It learns daily use and shows days left; a low LED/buzzer warns at 15%.

Product line / price in the plan: Tier B · low-voltage electronics · prototype Rs 1730 · price Rs 2499. Status: designed and verified as far as a computer can; no prototype built, nothing field-tested.

1. What you are building: deliverables

Deliverable	Where	State
PCB: Gerbers, BOM, CPL, KiCad project	`utility10/pcb/release/gastaraju/`	**order-ready: every automatic check passes** (DRC errors 0, netlist match True)
Firmware (`boards/gas_taraju.hex`)	`utility10/firmware/release/8MHz/boards/gas_taraju.hex`, source `utility10/firmware/src/main_board.cpp` (seeti: `seeti_prod.cpp`), drivers `board_drivers.h`, engine `utility10/firmware/lib/utilcore.c`	engine host-tested (115 checks); board firmware compiles but is **not yet run on hardware** (see section 6)
Enclosure STL (prototype shell)	`utility10/02-gas-taraju/enclosure_body.stl`, `enclosure_lid.stl` and the board's `4_enclosure/`	prototype shell only
Mechanical / bought-in parts	section 3	listed
Test sheet	`manuals/templates/eol_test_sheet.md`	template

2. Before you start

- Read `manuals/STANDARD_PROCEDURES.md` (sections A-F, J, K).
- Tools: soldering iron, flux, multimeter, USBasp programmer + avrdude, calipers, 3D printer or print service, bench supply with current limit.
- Power source for this product: **3 x AA**.

3. Bill of materials

3a. Circuit board parts (32 parts, 3 test points)

SMD parts marked `SMT (JLC)` are assembled by the fab from the JLC BOM; parts marked `hand/THT` are soldered by you. Every LCSC number was checked against an LCSC/JLCPCB listing on 2026-11-01; re-check stock on the day you order.

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
C1	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C2	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C3	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
R7	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	ZHOURI 2.54-2*3	C5116479	hand/THT
J1	BATT 3xAA	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
SW1	Power	SW_SPDT_PCM12	C&K PCM12SMTR	C221841	SMT (JLC)
Q2	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
C4	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
BZ1	Buzzer	Buzzer_12x9.5RM7.6	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869	hand/THT
Q1	MMBT3904	SOT-23	MMBT3904	C20526	SMT (JLC)
D2	1N4148WS	D_SOD-323	1N4148WS	C2128	SMT (JLC)
R6	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
U4	HX711	SOIC-16_3.9x9.9mm_P1.27mm	Avia HX711 SOP-16	C43656	SMT (JLC)
R14	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
C10	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C11	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C12	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C13	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R15	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
C14	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C15	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
J3	Load cell bridge E+ E- A+ A-	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
J4	TM1637 4-digit	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
SW2	Tare / zero	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
D1	Green	LED_0603_1608Metric	Everlight 19-217/GHC-YR1S2/3T (LCSC showed it out of stock on the search day: check)	C72043	SMT (JLC)
R5	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
D5	Amber	LED_0603_1608Metric	Everlight 19-213/Y2C-CQ2R2L/3T yellow 591 nm (marked obsolete by some distributors: check)	C72038	SMT (JLC)
R8	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
D6	Red	LED_0603_1608Metric	Everlight 19-217/R6C-AL1M2VY/3T (some distributors list it as discontinued: check)	C72044	SMT (JLC)
R9	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)

3b. Bought-in and mechanical items (estimates for about 2,000 units; replace with quotes)

Item	Rs
4 × 50 kg load cells	360
HX711 amp	15
MCU	45
3-digit display + LEDs	40
Laser-cut plate + frame (MDF/acrylic)	120
PCB	30
Buzzer	12
Battery/USB	30
Packaging	45
Assembly + calibration	60

4. Step 1: fabricate the PCB

Upload `utility10/pcb/release/gastaraju/1_fabrication/gas_taraju_gerbers.zip` with the settings in STANDARD_PROCEDURES section A. Finish: HASL lead-free. Board size and layer list are in `utility10/pcb/release/gastaraju/README.md`.

5. Step 2: assemble the board

1. Order SMT assembly as in STANDARD_PROCEDURES section B (or hand-solder the SMD parts in section D order).
2. Solder the through-hole parts listed `hand/THT` in 3a.
3. Clean, inspect under magnification, and compare with `2_assembly/*_assembly_top.pdf`.

6. Step 3: program the microcontroller

1. Connect a USBasp to the 6-pin ISP header J2 (pin 1 = square pad) with the board powered (3 x AA).
2. Set fuses: `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m` (internal 8 MHz RC, CKDIV8 off, brown-out 2.7 V (the 8 MHz speed-grade limit): the unit resets cleanly when the battery is flat).
3. Flash and verify: `avrdude -p m328p -c usbasp -U flash:w:utility10/firmware/release/8MHz/boards/gas_taraju.hex:i` then `-U flash:v:utility10/firmware/release/8MHz/boards/gas_taraju.hex:i`.
4. Remove the programmer, power-cycle the board, and confirm the start-up behaviour (section 9).

If you change the supply type (for example USB instead of batteries) use the hex built for that clock (`4MHz` for 2 x AAA, `8MHz` for every other tool).

Firmware pin map (generated from the PCB spec, so firmware and board agree):

Arduino pin	ATmega pin	Net
D3	1	HX_SCK
D5	9	LED_G
D6	10	LED_A
D7	11	LED_R
D8	12	TARE
D9	13	BUZ
A4	27	DIO
A5	28	DCLK
D2	32	HX_DOUT

What is verified: the control logic (`utilcore.c`) passes 115 host tests. **What is not:** the display, OLED, RTC and HX711 drivers and the button flows in `main_board.cpp` were written from datasheets and only compile-checked. Budget bring-up time for them on the first prototype.

7. Step 4: electrical bring-up

1. Bridge disconnected: DOUT idles high; with a 20 kg reference weight reading is stable within 0.1 kg after calibration.
2. Zero with the plate empty (tare button).

8. Step 5: sensors, cables and mechanical assembly

- **Plate and load cells.** Laser-cut the 330 mm plate from the SVG/DXF in `utility10/out/` (check the product README for the file name); fit four load cells (half-bridge or full-bridge, wiring to the HX711

E+/E-/A+/A- inputs) under the plate corners on a rigid, level base with four rubber feet. A warped base or a plate that rocks destroys accuracy.
- Bonded-load-cell wiring: keep cable lengths equal and short; shield away from the buzzer lead.

9. Step 6: enclosure and final assembly

1. Print the shell (STANDARD_PROCEDURES section G). Enclosure: `enclosure_body.stl` / `enclosure_lid.stl` (90×60×26 mm, 2.4 mm walls, watertight): a simple prototype shell; production tooling needs a proper design.
2. Place the board on the standoffs (4 mm), fasten with M2.5 screws (6 mm), route cables through the cut-outs, close the lid.
3. Label: serial number, rev, battery type/polarity, warnings.

10. Step 7: calibration

- Per-unit calibration is mandatory. Zero the empty plate, place a certified 20 kg reference weight, record counts per kg and offset, store in EEPROM; repeat at 5, 10, 15 kg to check linearity (error under +-0.1 kg). The user enters the cylinder tare weight stamped on the collar.

11. Step 8: end-of-line functional test (every unit)

1. Zero the plate; place a 20 kg reference weight: reading within +-0.1 kg; remove: returns to 0 +-0.05 kg.
2. Place a 14.2 kg net cylinder-equivalent (use test weights): reads 100 %; check the LED/buzzer low warning at 15 %.

Record results on `manuals/templates/eol_test_sheet.md`. Accept only if every line passes.

12. Packaging, documents, compliance

- In the box: unit, batteries or note on the required adapter, mounting accessories, quick-start card (`manuals/templates/quick_start_card.md`), warranty card.
- Compliance items to verify before sale: Battery safety, BIS requirement for any bought-in adapter, EMC, e-waste EPR, Legal Metrology labelling; no safety claims..
- Labelling: STANDARD_PROCEDURES K.

13. Troubleshooting (product specific)

Symptom	Cause / action
Reading drifts	plate not rigid, one load cell loaded unevenly, or HX711 supply noisy: check the 4V5 rail ripple
This is not a leak detector	do not describe it as a gas safety device: it is a weight scale only

General issues: STANDARD_PROCEDURES section L.

14. Known limits and risks

Host tests (110 checks incl. other tools): calibration, rejects readings while the cylinder is being placed or bumped, full cylinder reads 14.2 kg / 100%, learned consumption within 0.05 kg/day, empty reads 0%. A real bug was found: the sample history froze after 255 samples (index stuck), fixed. Not verified: real load cell drift/temperature, hysteresis under 30 kg, creep.

Accuracy needs calibration per unit; load cell cost drives BOM; cylinders have different tare weights (user enters it).

Nothing in this manual replaces testing with real users, a design review by a PCB engineer, and the compliance checks above.